Table 7.3 Rate constant for isomerisation

Rate mol $L^{-1}min^{-1}$	[cyclopropane] mol L^{-1}	$k = \frac{\text{Rate}}{[\text{cyclopropane}]}$
6.92×10^{-2}	2	3.46×10^{-2}
3.46×10^{-2}	1	3.46×10^{-2}
1.73×10^{-2}	0.5	3.46×10^{-2}

Let us consider an another example, the oxidation of nitric oxide (NO)

$$2NO(g) + O_2(g) \longrightarrow 2NO_2\ (g)$$

Series of experiments are conducted by keeping the concentration of one of the reactants constant and the changing the concentration of the others.

Experiment	[NO] X 10^{-2} (mol L^{-1})	[O_2] X 10^{-2} (mol L^{-1})	Initial rate x 10^{-2} (mol $L^{-1}min^{-1}$)
1	1.3	1.1	19.26
2	1.3	2.2	38.40
3	2.6	1.1	76.80

$$\text{Rate} = k\,[NO]^m[O_2]^n$$

For experiment 1, the rate law is

$$\text{Rate}_1 = k\,[NO]^m[O_2]^n$$

$$19.26\ X10^{-2} = k\,[1.3]^m[1.1]^n \quad ...(1)$$

Similarly for experiment 2

$$\text{Rate}_2 = k\,[NO]^m[O_2]^n$$

$$38.40\ X10^{-2} = k\,[1.3]^m[2.2]^n \quad ...(2)$$

For experiment 3

$$\text{Rate}_3 = k\,[NO]^m[O_2]^n$$

$$76.8 \times 10^{-2} = k\,[2.6]^m[1.1]^n \quad ...(3)$$

$$\frac{(2)}{(1)} \Rightarrow \frac{38.40 \times 10^{-2}}{19.26 \times 10^{-2}} = \frac{k\,[1.3]^m[2.2]^n}{k\,[1.3]^m[1.1]^n}$$

$$2 = \left(\frac{2.2}{1.1}\right)^n$$

$2=2^n$ i.e., n=1

Therefore the reaction is first order with respect to O_2

$$\frac{(3)}{(1)} \Rightarrow \frac{76.8 \times 10^{-2}}{19.26 \times 10^{-2}} = \frac{k\,[2.6]^m[1.1]^n}{k\,[1.3]^m[1.1]^n}$$

$$4 = \left(\frac{2.6}{1.3}\right)^m$$

$4=2^m$ i.e., m=2

Therefore the reaction is second order with respect to NO

The rate law is $\text{Rate}_1 = k\,[NO]^2[O_2]^1$

The overall order of the reaction = (2 + 1) = 3

Differences between rate and rate constant of a reaction:

s.no	Rate of a reaction	Rate constant of a reaction
1	It represents the speed at which the reactants are converted into products at any instant.	It is a proportionality constant

XII U7 kinetics - Jerald Folder.indd 209 22-12-2022 19:53:51